Asymptotes and Intercepts

Index

Key notes Page 2

Checked conventions Page 4

Finding asymptotes Page 5

Intercepts, domain, range and sketching Page 9

Finding parameters from graphs and asymptotes Page 21

Transformations and limits Page 31

Exam-style extended questions Page 37

Asymptotes & Intercepts

Rational Functions – Key Notes and Formulae

---

1. Vertical asymptote (V.A.)

Set the **denominator equal to zero** and solve. e.g. $f(x) = \dfrac{2x+7}{x-5}$: $x-5=0 \Rightarrow$ V.A. $x=5$.

2. Horizontal asymptote (H.A.) – compare the degree of the Numerator (N) and Denominator (D):

Case	Example	Horizontal asymptote
N < D	$\dfrac{2x}{3x^2+1}$	$y=0$
N = D	$\dfrac{2x^2}{3x^2+1}$	$y = \dfrac{\text{leading coeff. of N}}{\text{leading coeff. of D}} = \dfrac{2}{3}$
N > D (by 1)	$\dfrac{x^2+2x+1}{x}$	No H.A. – an **oblique** asymptote

Oblique asymptote: divide. $\dfrac{x^2+2x+1}{x} = x+2+\dfrac{1}{x}$, so the oblique asymptote is $y=x+2$.

3. Standard forms (IB)

- $y = \dfrac{ax+b}{cx+d}$: V.A. $x=-\dfrac{d}{c}$, H.A. $y=\dfrac{a}{c}$.
- $y = a+\dfrac{b}{cx+d}$: V.A. $x=-\dfrac{d}{c}$, H.A. $y=a$. (also $y=\dfrac{k}{x-h}+v$: V.A. $x=h$, H.A. $y=v$)
- To change $\dfrac{ax+b}{cx+d}$ into $A+\dfrac{B}{cx+d}$, divide: e.g. $\dfrac{2x+1}{x-3} = \dfrac{2(x-3)+7}{x-3} = 2+\dfrac{7}{x-3}$.

4. Intercepts

- x**-intercept:** set $y=0$ (numerator $=0$) and solve for x. Answer as a point $(x,0)$.
- y**-intercept:** set $x=0$ and find y. Answer as a point $(0,y)$.
- If the numerator is a constant (e.g. $\frac{5}{x+1}$), there is **no** x-intercept.

5. Domain and range (for $y=\frac{ax+b}{cx+d}$ or $y=a+\frac{b}{cx+d}$)

Domain: $x \neq$ V.A. **Range:** $y \neq$ H.A.

Key notes

6. Behaviour near asymptotes (limit notation)

For $f(x) = 2 + \dfrac{3}{x-1}$:

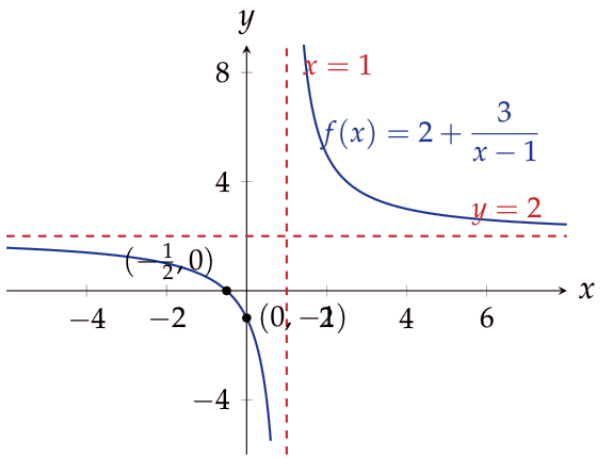
as $x \to 1^-$, $f(x) \to -\infty$ as $x \to -\infty$, $f(x) \to 2^-$ (from below)

as $x \to 1^+$, $f(x) \to +\infty$ as $x \to +\infty$, $f(x) \to 2^+$ (from above)

- $\lim\limits_{x \to \pm\infty} f(x) = k$ means $y = k$ is a horizontal asymptote.
- $\lim\limits_{x \to a^{\pm}} f(x) = \pm\infty$ means $x = a$ is a vertical asymptote.

7. Sketching a rational function – checklist

1. Draw both asymptotes as **dotted/dashed** lines and label their equations.
2. Plot and label the x- and y-intercepts.
3. The graph has two branches, in opposite "corners" formed by the asymptotes. Use the intercepts (or a test point) to decide which corners.
4. The curve gets closer and closer to each asymptote but never crosses the vertical one.



8. Transformations and asymptotes (for $y = f(x)$ with V.A. $x = p$ and H.A. $y = q$)

Transformation	H.A.	V.A.	Transformation	H.A.	V.A.
$f(x) + k$	$y = q + k$	$x = p$	$f(x - h)$	$y = q$	$x = p + h$
$a\,f(x)$	$y = aq$	$x = p$	$f(bx)$	$y = q$	$x = p/b$
$-f(x)$	$y = -q$	$x = p$	$f(-x)$	$y = q$	$x = -p$

Rule of thumb: changes *outside* f move the H.A.; changes *inside* the bracket move the V.A.

e.g. $\dfrac{1}{x} \to \dfrac{4}{x} \to \dfrac{4}{x-2} \to \dfrac{4}{x-2} + 3$: vertical stretch s.f. 4, translate 2 right, translate 3 up.

How to use this worksheet: questions go from easy to difficult. Attempt each page first, then check the yellow answer page that follows it.

Checked conventions

Vertical asymptotes and holes. Solve the original denominator equation to find excluded domain values. A common cancelled factor leaves a hole. Only a non-removable denominator zero produces a vertical asymptote.

Range. A genuine nonconstant linear-over-linear rational function excludes its horizontal-asymptote level; more general rational functions may cross a horizontal asymptote and may have additional range restrictions. Use the actual function and graph.

Inverse intersections. A function and its inverse are reflected in $y = x$, but their intersections are not universally restricted to that line. In Q14, solve the stated equality directly: $f(1) = 3 - a$ and $f^{-1}(1) = \frac{a+2}{4}$, so $3 - a = \frac{a+2}{4}$ gives $a = 2$.

Radical-domain endpoints (Q26). For $\frac{3}{\sqrt{9-x^2}}$, require $9 - x^2 > 0$: the domain is $-3 < x < 3$. Both endpoints are excluded and both are vertical asymptotes. The minimum is 1 at $x = 0$, so the range is $y \geq 1$.

Exact domains (Q28). The denominator roots are $\frac{1 \pm \sqrt{1+4\sqrt{2}}}{2}$, approximately -0.790044 and 1.790044. When specifying a domain, exclude the exact values rather than only their rounded approximations.

Question 1 Source worksheet · p.3

Question 1 Non-calculator Easy [Maximum mark: 18]

Complete the following table by writing down the corresponding asymptotes. The first row is done for you.

Function	Vertical asymptote	Horizontal asymptote
$f(x) = \frac{2x+7}{x-5}$	$x = 5$	$y = 2$
$f(x) = \frac{x+7}{2x-3}$		
$f(x) = \frac{-4x+1}{2x-6}$		
$f(x) = \frac{2}{x-5}$		
$f(x) = \frac{2}{x-5} + 1$		
$f(x) = \frac{2}{x-5} - 1$		
$f(x) = \frac{2x+7}{x-5} + 1$		
$f(x) = \frac{-4x+1}{2x-6} + 3$		
$f(x) = \frac{5-x}{5+x}$		
$f(x) = \frac{3x+134}{2x+5}$		

Answer 1 Source worksheet · p.3

Question 1

Function	Vertical asymptote	Horizontal asymptote
$f(x) = \dfrac{2x+7}{x-5}$	$x = 5$	$y = 2$
$f(x) = \dfrac{x+7}{2x-3}$	$x = \dfrac{3}{2}$	$y = \dfrac{1}{2}$
$f(x) = \dfrac{-4x+1}{2x-6}$	$x = 3$	$y = -2$
$f(x) = \dfrac{2}{x-5}$	$x = 5$	$y = 0$
$f(x) = \dfrac{2}{x-5} + 1$	$x = 5$	$y = 1$
$f(x) = \dfrac{2}{x-5} - 1$	$x = 5$	$y = -1$
$f(x) = \dfrac{2x+7}{x-5} + 1$	$x = 5$	$y = 2+1 = 3$
$f(x) = \dfrac{-4x+1}{2x-6} + 3$	$x = 3$	$y = -2+3 = 1$
$f(x) = \dfrac{5-x}{5+x}$	$x = -5$	$y = \dfrac{-1}{1} = -1$
$f(x) = \dfrac{3x+134}{2x+5}$	$x = -\dfrac{5}{2}$	$y = \dfrac{3}{2}$

Tip: V.A.: denominator $= 0$. H.A.: ratio of the x-coefficients; a "$+k$" outside the fraction moves the H.A. up by k.

Question 2 Source worksheet · p.5

Question 2 Non-calculator Easy [Maximum mark: 6]

Consider the graph of $f(x) = \frac{6x+1}{2x-4}$.

(a) Write down the domain and the range of f. [2]

(b) Write down the equation of (i) the horizontal asymptote; (ii) the vertical asymptote. [2]

(c) Find the x-intercept and the y-intercept. [2]

Question 3 Source worksheet · p.5

Question 3 Non-calculator Medium [Maximum mark: 10]

(a) Express the function $f(x) = 4 + \frac{5}{x+3}$ in the form $f(x) = \frac{ax+b}{x+3}$. [2]

(b) Express the function $g(x) = \frac{x+10}{x+3}$ in the form $g(x) = A + \frac{B}{x+3}$. [2]

(c) Express the function $h(x) = \frac{2x+10}{x+3}$ in the form $h(x) = C + \frac{D}{x+3}$. [2]

(d) Complete the following table. [4]

	f	g	h
Vertical asymptote			
Horizontal asymptote			

Answer 2 Source worksheet · p.5

Question 2

(a) V.A. is where $2x-4=0$, i.e. $x=2$; H.A. $y=\frac{6}{2}=3$.
Domain: $\boxed{x\neq 2}$, Range: $\boxed{y\neq 3}$

(b) (i) $\boxed{y=3}$ (ii) $\boxed{x=2}$

(c) x-intercept: $6x+1=0\Rightarrow x=-\frac{1}{6}$, so $\boxed{\left(-\frac{1}{6},0\right)}$

y-intercept: $f(0)=\dfrac{1}{-4}=-\dfrac{1}{4}$, so $\boxed{\left(0,-\frac{1}{4}\right)}$

Answer 3 Source worksheet · p.5

Question 3

(a) $f(x)=4+\dfrac{5}{x+3}=\dfrac{4(x+3)}{x+3}+\dfrac{5}{x+3}=\boxed{\dfrac{4x+17}{x+3}}$

(b) $g(x)=\dfrac{x+10}{x+3}=\dfrac{(x+3)+7}{x+3}=\boxed{1+\dfrac{7}{x+3}}$ $(A=1,\ B=7)$

(c) $h(x)=\dfrac{2x+10}{x+3}=\dfrac{2(x+3)+4}{x+3}=\boxed{2+\dfrac{4}{x+3}}$ $(C=2,\ D=4)$

(d)

	f	g	h
Vertical asymptote	$x=-3$	$x=-3$	$x=-3$
Horizontal asymptote	$y=4$	$y=1$	$y=2$

Question 4

Source worksheet · p.7

Question 4

Calculator optional | Easy

[Maximum mark: 10]

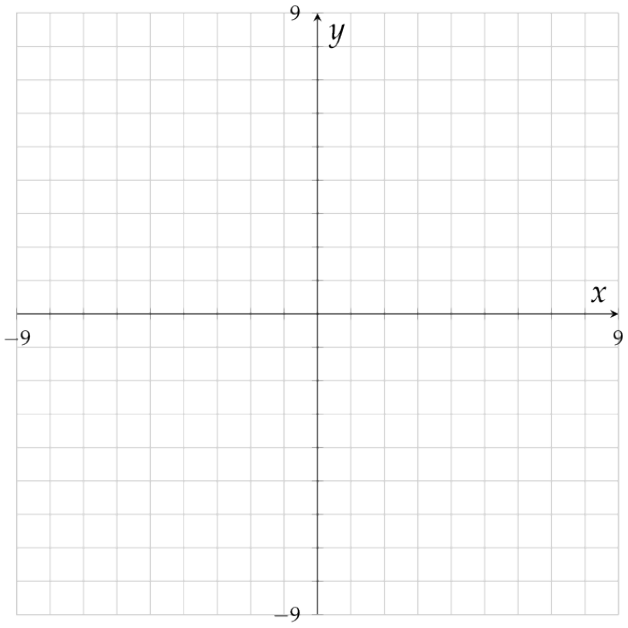
Let $f(x) = \dfrac{3}{x-2}$.

(a) Write down the equations of the asymptotes. [2]

(b) Find any x- or y-intercepts. [2]

(c) Sketch the graph of the function. Indicate the information found in (a) and (b). [4]

(d) Write down the domain and the range of the function. [2]



Answer 4 Source worksheet · p.7

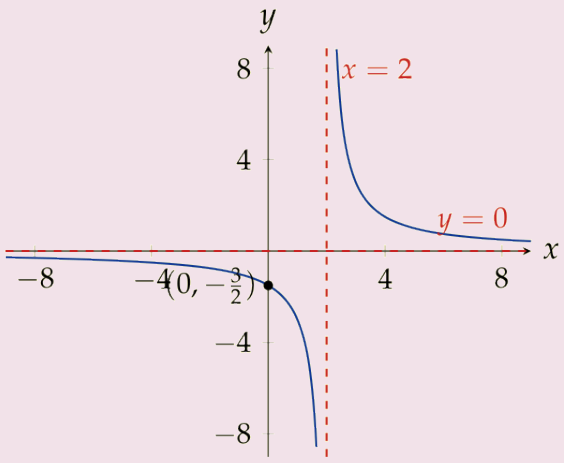
Question 4

(a) H.A.: $\boxed{y = 0}$ (numerator degree $<$ denominator degree), V.A.: $x - 2 = 0 \Rightarrow \boxed{x = 2}$

(b) x-intercept: $\dfrac{3}{x-2} = 0$ has no solution (numerator is never 0), so **there is no x-intercept**.

y-intercept: $f(0) = \dfrac{3}{-2}$, so $\boxed{(0, -\frac{3}{2})}$

(c)



(d) Domain: $\boxed{x \neq 2}$ Range: $\boxed{y \neq 0}$

Question 5 Source worksheet · p.9

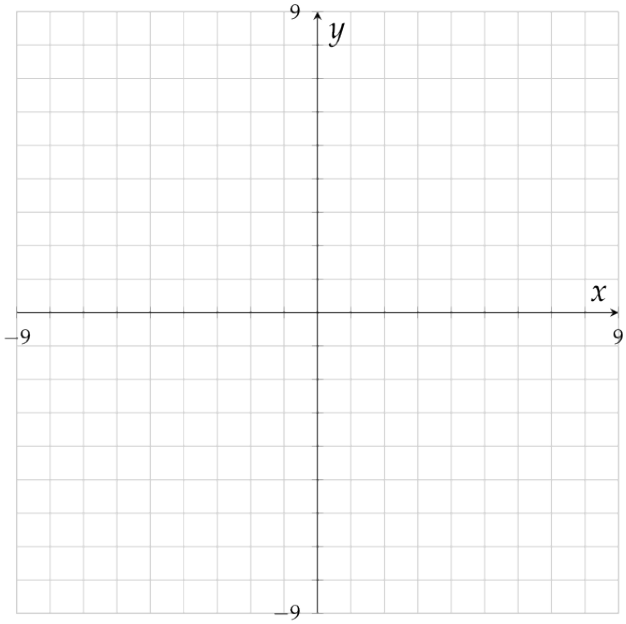
Question 5 Non-calculator Medium [Maximum mark: 8]

Let $f(x) = \dfrac{3x-6}{x+1}$, for $x \neq -1$.

(a) For the graph of $y = f(x)$, find the coordinates of:
(i) the x-intercept; (ii) the y-intercept. [3]

(b) For the graph of $y = f(x)$, find the equation of:
(i) the horizontal asymptote; (ii) the vertical asymptote. [3]

(c) Hence sketch the graph of $y = f(x)$. [2]



Answer 5 Source worksheet · p.9

Question 5

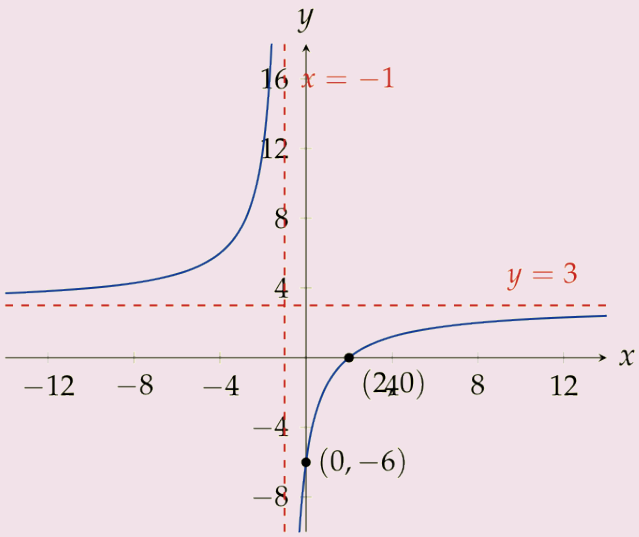
(a) (i) $f(x) = 0 \Rightarrow 3x - 6 = 0 \Rightarrow x = 2$, so $\boxed{(2, 0)}$

(ii) $f(0) = \dfrac{3(0) - 6}{0 + 1} = -6$, so $\boxed{(0, -6)}$

(b) (i) $\lim\limits_{x \to \pm\infty} f(x) = \dfrac{3}{1} = 3$, so H.A. $\boxed{y = 3}$

(ii) $x + 1 = 0 \Rightarrow$ V.A. $\boxed{x = -1}$

(c)



Question 6

Source worksheet · p.11

Question 6

Calculator optional | Medium

[Maximum mark: 12]

Let $f(x) = \dfrac{3x-6}{x+2}$.

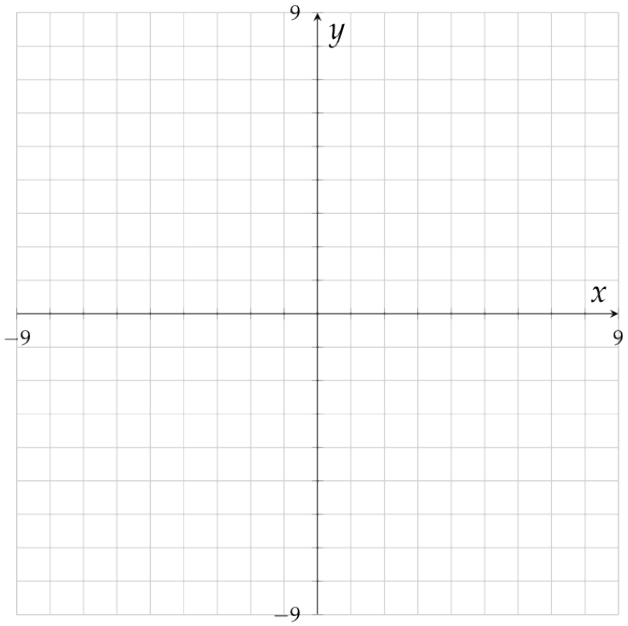
(a) Write down the equations of the asymptotes. [2]

(b) Find the x- and y-intercepts. [2]

(c) Sketch the graph of the function. Indicate the information found in (a) and (b). [4]

(d) Write down the domain and the range of the function. [2]

(e) Find the point of intersection between the graph of f and the line $y = 1$. [2]



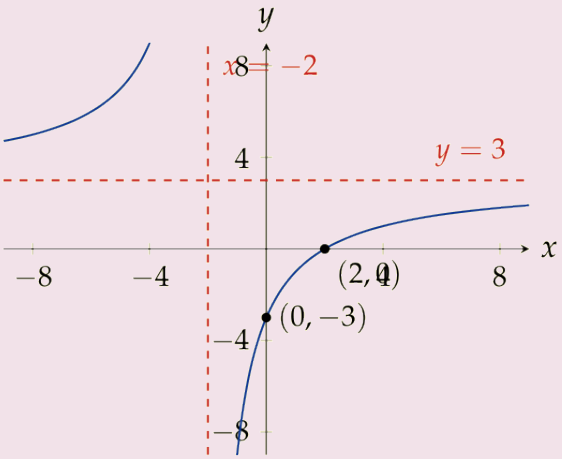
Answer 6 Source worksheet · p.11

Question 6

(a) H.A.: $\boxed{y = 3}$, V.A.: $\boxed{x = -2}$

(b) x-intercept: $3x - 6 = 0 \Rightarrow \boxed{(2, 0)}$; y-intercept: $f(0) = \frac{-6}{2} = -3 \Rightarrow \boxed{(0, -3)}$

(c)



(d) Domain: $\boxed{x \neq -2}$ Range: $\boxed{y \neq 3}$

(e) $\dfrac{3x - 6}{x + 2} = 1 \Rightarrow 3x - 6 = x + 2 \Rightarrow 2x = 8 \Rightarrow x = 4$, so the point is $\boxed{(4, 1)}$

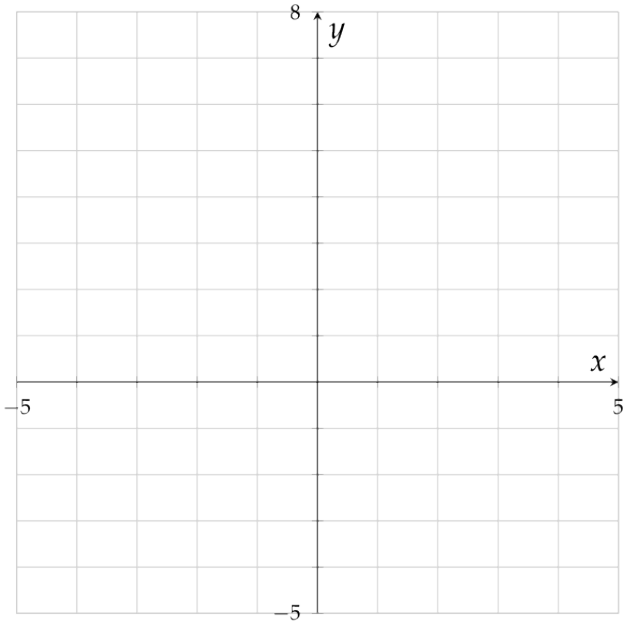
Question 7 Source worksheet · p.13

Question 7 Calculator optional Medium [Maximum mark: 7]

The function $f(x)$ is defined as $f(x) = 3 + \frac{1}{2x-5}$, $x \neq \frac{5}{2}$.

(a) Sketch the curve of f for $-5 \leq x \leq 5$, showing the asymptotes. [3]

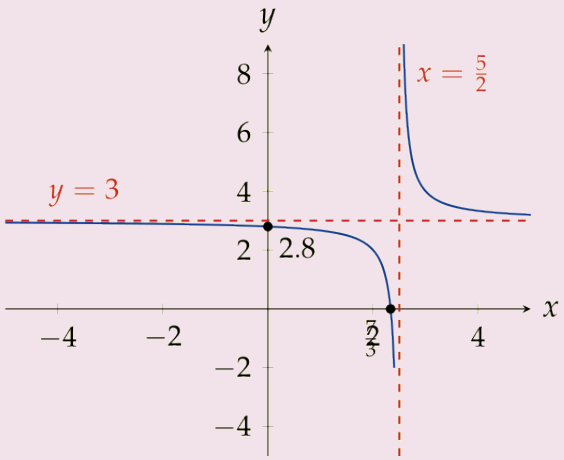
(b) Using your sketch, write down
(i) the equation of each asymptote; (ii) the value of the x-intercept; (iii) the value of the y-intercept. [4]



Answer 7 Source worksheet · p.13

Question 7

(a)



(b) (i) $\boxed{y = 3}$ and $\boxed{x = \frac{5}{2}}$ (must be equations)

(ii) $3 + \dfrac{1}{2x-5} = 0 \Rightarrow 2x - 5 = -\dfrac{1}{3} \Rightarrow x = \dfrac{14}{6} = \boxed{\dfrac{7}{3} \approx 2.33}$

(iii) $f(0) = 3 + \dfrac{1}{-5} = \boxed{\dfrac{14}{5} = 2.8}$

Question 8 Source worksheet · p.15

Question 8 Non-calculator Medium [Maximum mark: 12]

The function f is given by $f(x) = \dfrac{2x+1}{x-3}$, $x \in \mathbb{R}, x \neq 3$.

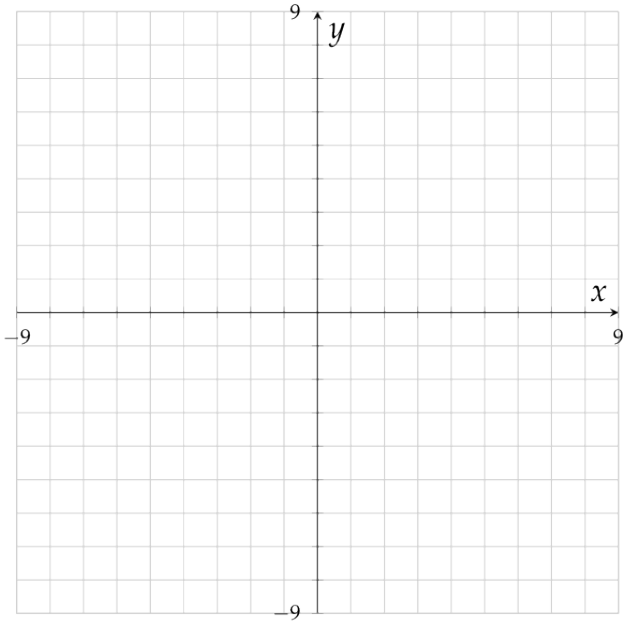
(a) (i) Show that $y = 2$ is an asymptote of the graph of $y = f(x)$. [2]

(ii) Find the vertical asymptote of the graph. [1]

(iii) Write down the coordinates of the point P at which the asymptotes intersect. [1]

(b) Find the points of intersection of the graph and the axes. [4]

(c) Hence sketch the graph of $y = f(x)$, showing the asymptotes by dotted lines. [4]



Answer 8 Source worksheet · p.15

Question 8

(a) (i) $f(x) = \dfrac{2x+1}{x-3} = \dfrac{2(x-3)+7}{x-3} = 2 + \dfrac{7}{x-3}$.

As $|x| \to \infty$, $\dfrac{7}{x-3} \to 0$, so $f(x) \to 2 \Rightarrow y = 2$ is an asymptote.

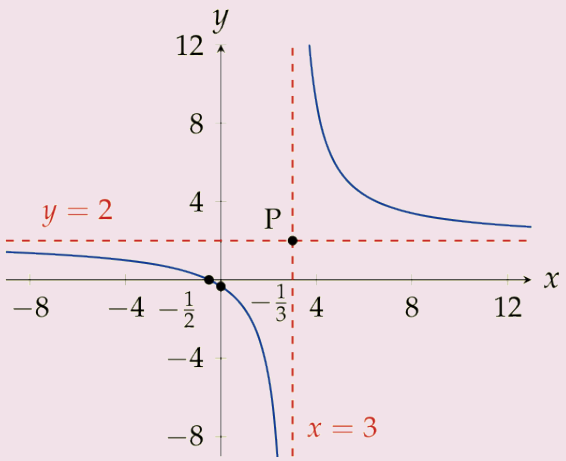
OR $\lim\limits_{x\to\infty} \dfrac{2x+1}{x-3} = 2$.

(ii) $\boxed{x = 3}$ (iii) $\boxed{P(3,\,2)}$

(b) $f(x) = 0 \Rightarrow 2x+1 = 0 \Rightarrow x = -\frac{1}{2}$: $\boxed{\left(-\frac{1}{2}, 0\right)}$

$x = 0 \Rightarrow f(0) = \dfrac{1}{-3}$: $\boxed{\left(0, -\frac{1}{3}\right)}$

(c)



Marks: asymptotes (1), intercepts (1), shape (2).

Question 9

Source worksheet · p.17

Question 9

Non-calculator Medium [Maximum mark: –]

For each of the following functions:

(i) Find the equations of the asymptotes. (ii) State the domain and range.

(iii) Find the axes intercepts.

(iv) Discuss the behaviour of the function as it approaches its asymptotes.

(v) Sketch the graph of the function.

(a) $f(x) = 2 + \frac{1}{x-3}$ (b) $f(x) = 2 - \frac{3}{x+1}$ (c) $f(x) = \frac{x}{x-1}$

Hint for (c): write $\frac{x}{x-1} = 1 + \frac{1}{x-1}$.

Working:

Answer 9 Source worksheet · p.17

Question 9

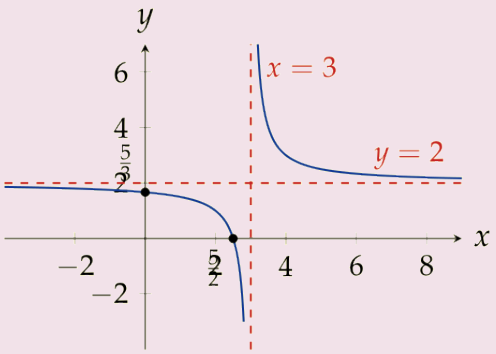
(a) $f(x) = 2 + \dfrac{1}{x-3}$

(i) V.A. $x = 3$, H.A. $y = 2$

(ii) Domain $x \neq 3$, Range $y \neq 2$

(iii) x-int $\frac{5}{2}$, y-int $\frac{5}{3}$

(iv) as $x \to 3^-$, $f(x) \to -\infty$;
as $x \to 3^+$, $f(x) \to +\infty$;
as $x \to -\infty$, $f(x) \to 2^-$;
as $x \to +\infty$, $f(x) \to 2^+$



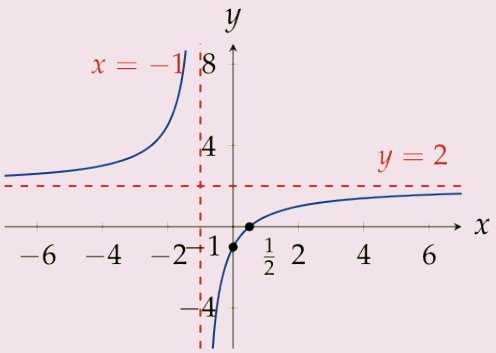
(b) $f(x) = 2 - \dfrac{3}{x+1}$

(i) V.A. $x = -1$, H.A. $y = 2$

(ii) Domain $x \neq -1$, Range $y \neq 2$

(iii) x-int $\frac{1}{2}$, y-int -1

(iv) as $x \to -1^-$, $f(x) \to +\infty$;
as $x \to -1^+$, $f(x) \to -\infty$;
as $x \to -\infty$, $f(x) \to 2^+$;
as $x \to +\infty$, $f(x) \to 2^-$



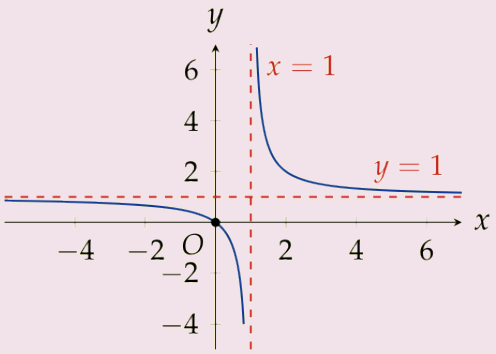
(c) $f(x) = \dfrac{x}{x-1} = 1 + \dfrac{1}{x-1}$

(i) V.A. $x = 1$, H.A. $y = 1$

(ii) Domain $x \neq 1$, Range $y \neq 1$

(iii) x-int 0, y-int 0 (passes through origin)

(iv) as $x \to 1^-$, $f(x) \to -\infty$;
as $x \to 1^+$, $f(x) \to +\infty$;
as $x \to -\infty$, $f(x) \to 1^-$;
as $x \to +\infty$, $f(x) \to 1^+$



Question 10

Source worksheet · p.19

Question 10

Non-calculator Easy [Maximum mark: 6]

The function f is of the form $f(x) = \frac{ax+b}{2x+c}$, for $x \neq -\frac{c}{2}$, where $a, b, c \in \mathbb{Z}$. Given that the graph of $y = f(x)$ has asymptotes $x = -5$ and $y = 2$, and that the point $\mathrm{P}\left(1, -\frac{1}{12}\right)$ lies on the graph, find the values of a, b and c.

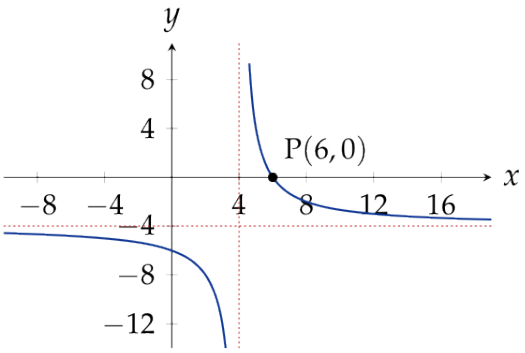
Question 11

Source worksheet · p.19

Question 11

Non-calculator Easy [Maximum mark: 4]

A rational function is defined by $f(x) = a + \frac{b}{x-c}$, for $x \neq c$, where $a, b, c \in \mathbb{Z}$. The following diagram represents the graph of $y = f(x)$.



Using the information on the graph,

(a) state the value of a and the value of c; [2]

(b) find the value of b. [2]

Answer 10 Source worksheet · p.19

Question 10

Vertical asymptote $x = -5$: denominator $= 0$ at $x = -5$:

$$2(-5) + c = 0 \Rightarrow \boxed{c = 10}$$

Horizontal asymptote $y = 2$: $\lim_{x \to \pm\infty} f(x) = \frac{a}{2} = 2 \Rightarrow \boxed{a = 4}$

Point P: $f(1) = -\frac{1}{12}$

$$\frac{4 + b}{2 + 10} = -\frac{1}{12} \Rightarrow 4 + b = -1 \Rightarrow \boxed{b = -5}$$

Answer 11 Source worksheet · p.19

Question 11

(a) From the graph, H.A. $y = -4$ and V.A. $x = 4$: $\boxed{a = -4}$ and $\boxed{c = 4}$

(b) P$(6, 0)$ lies on the graph, so $f(6) = 0$:

$$0 = -4 + \frac{b}{6 - 4} \Rightarrow 4 = \frac{b}{2} \Rightarrow \boxed{b = 8}$$

Question 12

Source worksheet · p.21

Question 12

Non-calculator Easy [Maximum mark: 5]

Consider the function $f(x) = b + \frac{4}{x+a}$, where $a, b \in \mathbb{Z}$.

The equation of the vertical asymptote of $f(x)$ is $x = 2$ and $f\left(\frac{2}{3}\right) = 0$.

(a) Write down the value of a. [1]

(b) Hence, find the value of b. [3]

(c) Hence, write down the equation of the horizontal asymptote. [1]

Question 13

Source worksheet · p.21

Question 13

Non-calculator Medium [Maximum mark: 6]

Let $f(x) = \frac{2}{x-a} + b$, for $x \neq a$. The line $x = 1$ is a vertical asymptote of the graph of $y = f(x)$.

(a) Write down the value of a. [1]

The graph of $y = f(x)$ passes through the point $\mathrm{P}(0, 3)$.

(b) Find the value of b. [3]

(c) Find $\lim_{x \to \infty} f(x)$. [2]

Working:

Answer 12 Source worksheet · p.21

Question 12

(a) The V.A. is where $x + a = 0$. Since the V.A. is $x = 2$: $2 + a = 0 \Rightarrow \boxed{a = -2}$

(b) $\left(\frac{2}{3}, 0\right)$ lies on the curve:

$$0 = b + \frac{4}{\frac{2}{3} - 2} = b + \frac{4}{-\frac{4}{3}} = b - 3 \;\Rightarrow\; \boxed{b = 3}$$

(c) As $x \to \pm\infty$, $\dfrac{4}{x-2} \to 0$, so the H.A. is $\boxed{y = 3}$ (must be an equation, $y = 3$ or $f(x) = 3$).

Answer 13 Source worksheet · p.21

Question 13

(a) V.A. $x = 1$, so $\boxed{a = 1}$

(b) $f(0) = 3$: $\dfrac{2}{0-1} + b = 3 \;\Rightarrow\; -2 + b = 3 \;\Rightarrow\; \boxed{b = 5}$

(c) $\lim\limits_{x\to\infty} f(x) = \lim\limits_{x\to\infty}\left[\dfrac{2}{x-1} + 5\right] = 0 + 5 = \boxed{5}$

Question 14 Source worksheet · p.23

Question 14 Non-calculator Easy [Maximum mark: 7]

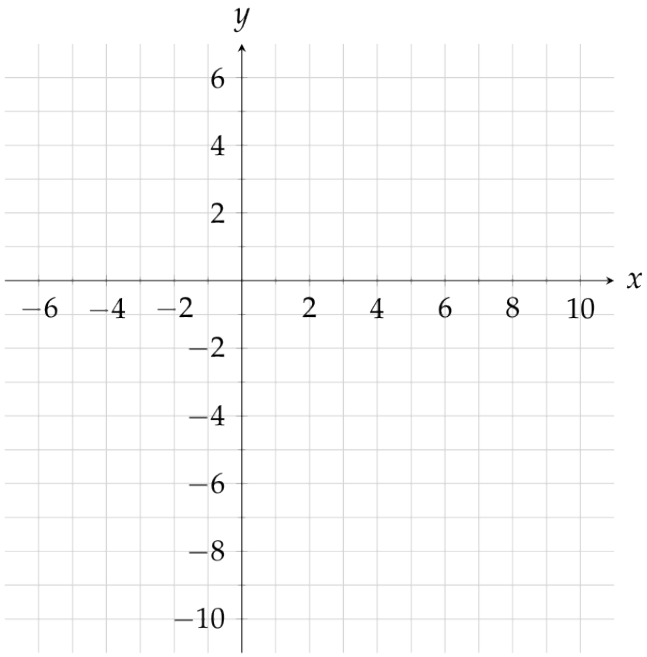
A function f is defined by $f(x) = \dfrac{a - 3x}{x - 2}$, where $x \in \mathbb{R}$, $x \neq 2$ and $a \in \mathbb{Z}$.

(a) For the graph of $y = f(x)$, write down the equation of (i) the vertical asymptote; (ii) the horizontal asymptote. [2]

It is known that $f(1) = f^{-1}(1)$.

(b) Find the value of a. [3]

(c) Hence, sketch the graph of f on the axes below. On your sketch, indicate the values of any axis intercepts and label any asymptotes with their equations. [2]



Answer 14 Source worksheet · p.23

Question 14

(a) (i) $\boxed{x = 2}$ (ii) $\lim_{x \to \pm\infty} \dfrac{-3x}{x} = -3$, so $\boxed{y = -3}$

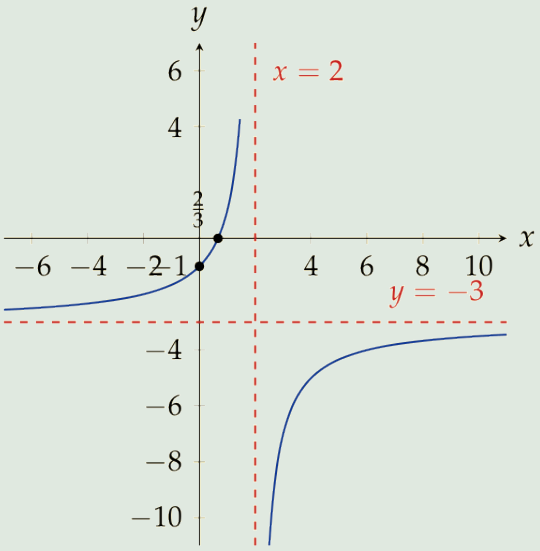
$$\frac{a - 3(1)}{1 - 2} = 1 \;\Rightarrow\; -a + 3 = 1 \;\Rightarrow\; \boxed{a = 2}$$

(c) $f(x) = \dfrac{2 - 3x}{x - 2}$

y-intercept: $f(0) = \dfrac{2}{-2} = -1$

x-intercept: $2 - 3x = 0 \Rightarrow x = \dfrac{2}{3}$

Marks: intercepts and asymptotes (1), shape (1).



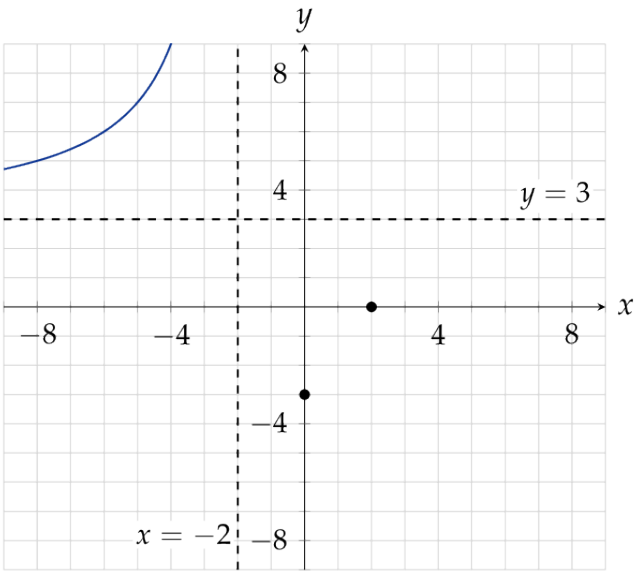
Checked proof: $f^{-1}(x) = \frac{a+2x}{x+3}$ for $a \neq 6$. Therefore $3 - a = \frac{a+2}{4}$ gives $12 - 4a = a + 2$, hence $a = 2$. No universal inverse-intersection rule is assumed.

Question 15 Source worksheet · p.25

Question 15

Non-calculator Medium [Maximum mark: 8]

The graph of the function $f(x) = \dfrac{ax+b}{x+c}$ is shown below.



(a) Find the values of a, b, c. [4]

(b) Find $f(4)$ and $f^{-1}(1)$. [2]

(c) By observing the graph, solve the inequality $f(x) \leq 1$. [2]

Answer 15

Source worksheet · p.25

Question 15

(a) V.A. $x = -2 \Rightarrow \boxed{c = 2}$. H.A. $y = \frac{a}{1} = 3 \Rightarrow \boxed{a = 3}$.

y-intercept $(0, -3)$: $\frac{b}{c} = \frac{b}{2} = -3 \Rightarrow \boxed{b = -6}$ (check: x-int $3x - 6 = 0 \Rightarrow x = 2$ ✓)

So $f(x) = \dfrac{3x - 6}{x + 2}$.

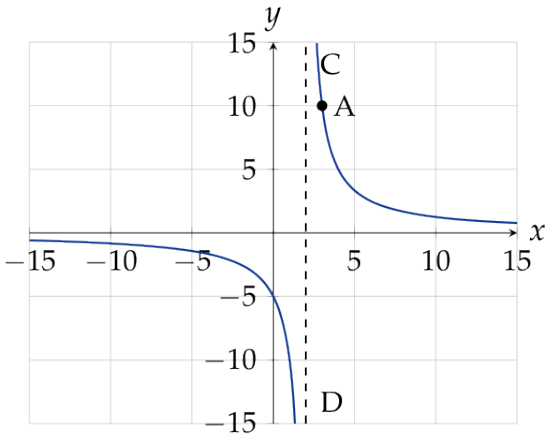
(b) $f(4) = \dfrac{12 - 6}{6} = \boxed{1}$. Hence $f(4) = 1$ means $\boxed{f^{-1}(1) = 4}$.

(c) The graph is on or below $y = 1$ between the asymptote and $x = 4$: $\boxed{-2 < x \leq 4}$

Question 16 Source worksheet · p.27

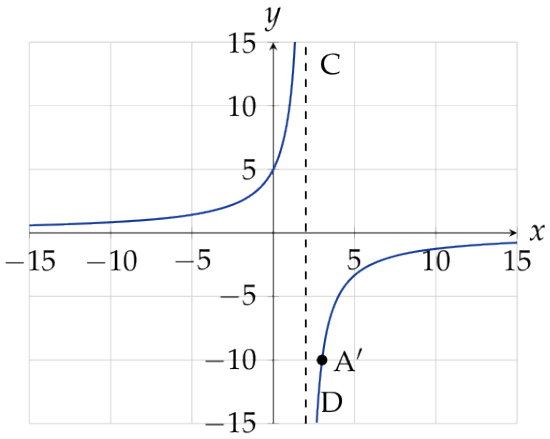
Question 16 Calculator optional Medium [Maximum mark: 6]

(a) The diagram shows part of the graph of the function $f(x) = \frac{q}{x-p}$. The curve passes through the point A$(3, 10)$. The line (CD) is an asymptote.



Find the value of (i) p; (ii) q. [4]

(b) The graph of $f(x)$ is transformed as shown in the following diagram. The point A is transformed to A$'(3, -10)$.



Give a full geometric description of the transformation. [2]

Answer 16 Source worksheet · p.27

Question 16

(a) (i) The asymptote (CD) is the line $x = 2$, so $\boxed{p = 2}$.

(ii) A$(3, 10)$ is on the graph: $10 = \dfrac{q}{3-2} \Rightarrow \boxed{q = 10}$

(b) The point $(3, 10)$ goes to $(3, -10)$: the y-values change sign.
The transformation is a **reflection in the x-axis**. (new function $y = -f(x) = -\dfrac{10}{x-2}$)

Question 17

Source worksheet · p.29

Question 17

Non-calculator Medium [Maximum mark: 6]

Let $g(x) = 3 + \frac{4}{x-2}$.

(a) Write down the equations of the asymptotes of the graph of g. [2]

(b) Describe the sequence of transformations that map the graph of $f(x) = \frac{1}{x}$ into the graph of g. [4]

Question 18

Source worksheet · p.29

Question 18

Non-calculator Hard [Maximum mark: 7]

Let $h(x) = \frac{3x+1}{x-2}$.

(a) Write down the equations of the asymptotes of the graph of h. [2]

(b) Describe the sequence of transformations that map the graph of $f(x) = \frac{1}{x}$ into the graph of h. [5]

Hint: first write $h(x)$ in the form $A + \frac{B}{x-2}$.

Working:

Answer 17 Source worksheet · p.29

Question 17

(a) H.A.: $\boxed{y = 3}$, V.A.: $\boxed{x = 2}$

(b) $\dfrac{1}{x} \longrightarrow \dfrac{4}{x} \longrightarrow \dfrac{4}{x-2} \longrightarrow \dfrac{4}{x-2} + 3$

- Vertical stretch with scale factor 4
- Horizontal translation 2 units to the right
- Vertical translation 3 units up (i.e. translation by vector $\binom{2}{3}$)

Answer 18 Source worksheet · p.29

Question 18

(a) H.A.: $\boxed{y = 3}$, V.A.: $\boxed{x = 2}$

(b) First rewrite: $h(x) = \dfrac{3x+1}{x-2} = \dfrac{3(x-2)+7}{x-2} = 3 + \dfrac{7}{x-2}$

$$\frac{1}{x} \longrightarrow \frac{7}{x} \longrightarrow \frac{7}{x-2} \longrightarrow \frac{7}{x-2} + 3$$

- Vertical stretch with scale factor 7
- Horizontal translation 2 units to the right
- Vertical translation 3 units up

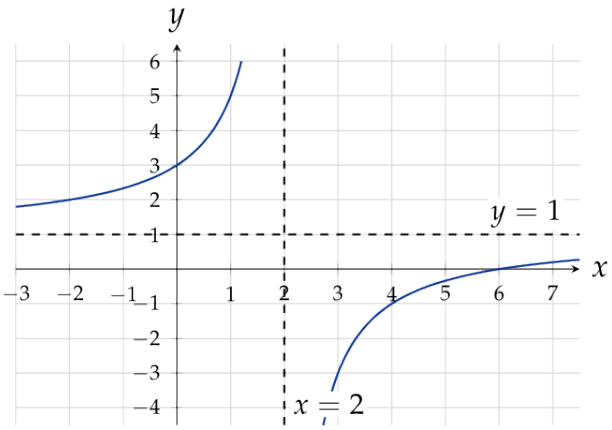
Question 19

Source worksheet · p.31

Question 19

Non-calculator Medium [Maximum mark: 19]

The graph of the function $f(x) = \dfrac{x-6}{x-2}$ is shown below.



(a) Write down the images of the horizontal asymptote $y = 1$ under the following transformations. [9]

$y = f(x) + 2$	$y = 3$	$y = f(x+2)$	
$y = f(x) - 2$		$y = f(x-2)$	
$y = 2f(x)$		$y = f(2x)$	
$y = f(x)/2$		$y = f(x/2)$	
$y = -f(x)$		$y = f(-x)$	

(b) Write down the images of the vertical asymptote $x = 2$ under the following transformations. [10]

$y = f(x) + 2$		$y = f(x+2)$	
$y = f(x) - 2$		$y = f(x-2)$	
$y = 2f(x)$		$y = f(2x)$	
$y = f(x)/2$		$y = f(x/2)$	
$y = -f(x)$		$y = f(-x)$	

Answer 19 Source worksheet · p.31

Question 19

(a) Images of the horizontal asymptote $y = 1$ (only changes *outside* f affect it):

$y = f(x) + 2$	$y = 3$	$y = f(x + 2)$	$y = 1$
$y = f(x) - 2$	$y = -1$	$y = f(x - 2)$	$y = 1$
$y = 2f(x)$	$y = 2$	$y = f(2x)$	$y = 1$
$y = f(x)/2$	$y = \frac{1}{2}$	$y = f(x/2)$	$y = 1$
$y = -f(x)$	$y = -1$	$y = f(-x)$	$y = 1$

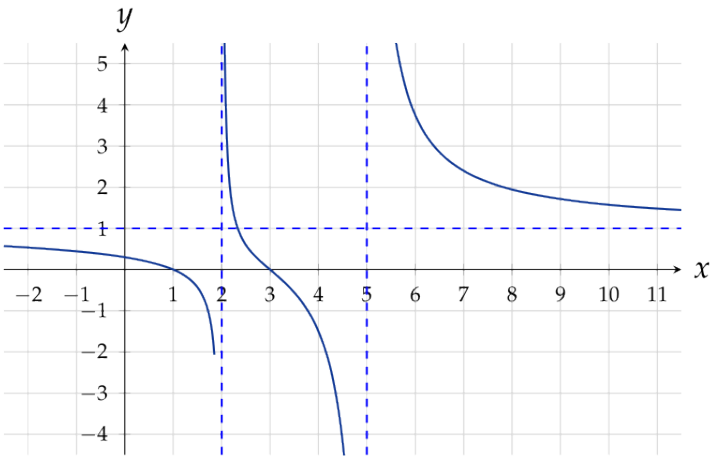
(b) Images of the vertical asymptote $x = 2$ (only changes *inside* the bracket affect it):

$y = f(x) + 2$	$x = 2$	$y = f(x + 2)$	$x = 0$
$y = f(x) - 2$	$x = 2$	$y = f(x - 2)$	$x = 4$
$y = 2f(x)$	$x = 2$	$y = f(2x)$	$x = 1$
$y = f(x)/2$	$x = 2$	$y = f(x/2)$	$x = 4$
$y = -f(x)$	$x = 2$	$y = f(-x)$	$x = -2$

Question 20 Source worksheet · p.33

Question 20 Non-calculator Medium [Maximum mark: 7]

The graph of a function f is shown below.



(a) Write down the equation of the horizontal asymptote. [1]

(b) Write down the equations of all vertical asymptotes. [1]

(c) Complete the following table. [3]

(1) $\lim_{x\to+\infty} f(x) =$	(4) $\lim_{x\to-\infty} f(x) =$
(2) $\lim_{x\to2^-} f(x) =$	(5) $\lim_{x\to2^+} f(x) =$
(3) $\lim_{x\to5^-} f(x) =$	(6) $\lim_{x\to5^+} f(x) =$

(d) State the results from part (c) which justify (i) the horizontal asymptote; (ii) the vertical asymptotes. [2]

Answer 20

Source worksheet · p.33

Question 20

(a) $\boxed{y = 1}$

(b) $\boxed{x = 2}$ and $\boxed{x = 5}$

(c)

(1) $\lim_{x \to +\infty} f(x) = 1$	(4) $\lim_{x \to -\infty} f(x) = 1$
(2) $\lim_{x \to 2^-} f(x) = -\infty$	(5) $\lim_{x \to 2^+} f(x) = +\infty$
(3) $\lim_{x \to 5^-} f(x) = -\infty$	(6) $\lim_{x \to 5^+} f(x) = +\infty$

(d) (i) Results (1) and (4) justify the horizontal asymptote $y = 1$.
(ii) Results (2) and (5) justify $x = 2$; results (3) and (6) justify $x = 5$.

Question 21 Source worksheet · p.35

Question 21 Non-calculator Medium [Maximum mark: 15]

Let $f(x) = \dfrac{5}{x+1}$, for $x \neq -1$.

(a) For the graph of f, find: (i) the x-intercept; (ii) the y-intercept; (iii) the equation of the vertical asymptote. [5]

Let $g(x) = x - 3$, for $x \in \mathbb{R}$. The graphs of f and g intersect at points A and B.

(b) Find the coordinates of A and B. [5]

(c) Find the equation of the straight line that passes through A and B, giving your answer in the form $y = mx + c$. [3]

(d) Write down the gradient of the line that is perpendicular to the line passing through A and B. [2]

Answer 21 Source worksheet · p.35

Question 21

(a) (i) $\dfrac{5}{x+1} = 0$ has no solutions, so there are **no x-intercepts**.

(ii) $f(0) = \dfrac{5}{0+1} = 5$, so the y-intercept is $\boxed{(0,5)}$

(iii) $x+1=0 \Rightarrow \boxed{x=-1}$

(b) Solve $f(x) = g(x)$:

$$\frac{5}{x+1} = x-3 \Rightarrow 5 = (x+1)(x-3) = x^2-2x-3 \Rightarrow x^2-2x-8=0 \Rightarrow (x+2)(x-4) =$$

$x=-2$ or $x=4$. Then $g(-2)=-5$, $g(4)=1$. $\boxed{\text{A}(-2,-5)}$ and $\boxed{\text{B}(4,1)}$

(c) $m_{\text{AB}} = \dfrac{1-(-5)}{4-(-2)} = \dfrac{6}{6} = 1$

$y-(-5) = 1\,(x-(-2)) \Rightarrow y+5 = x+2 \Rightarrow \boxed{y=x-3}$

(d) $m_{\perp} = -\dfrac{1}{m_{\text{AB}}} = \boxed{-1}$

Question 22 Source worksheet · p.37

Question 22 Calculator Medium [Maximum mark: 17]

Let $f(x) = \dfrac{3x+4}{2x-5}$, for $x \neq \dfrac{5}{2}$.

(a) For the graph of $y = f(x)$, find the coordinates of: (i) the x-intercept; (ii) the y-intercept. [3]

(b) For the graph of $y = f(x)$, find the equation of: (i) the horizontal asymptote; (ii) the vertical asymptote. [3]

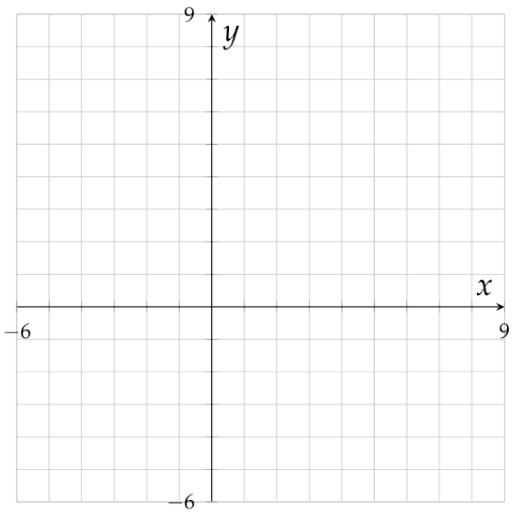
(c) Hence sketch the graph of $y = f(x)$. [2]

(d) Find $f^{-1}(x)$. [3]

The graphs of $f(x)$ and $f^{-1}(x)$ have two points of intersection.

(e) Find the coordinates of the two points of intersection. **Give your answers to two decimal places.** [3]

(f) Hence find the equation of the perpendicular bisector to the two points of intersection. [3]



Answer 22 Source worksheet · p.37

Question 22

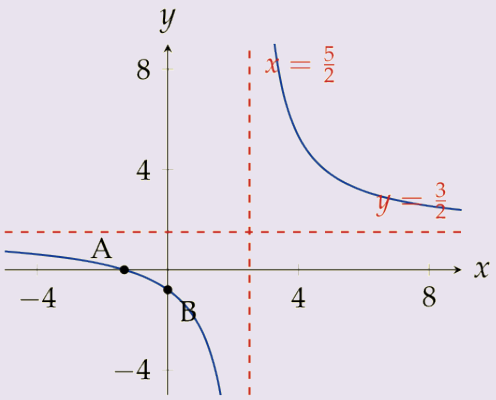
(a) (i) $3x+4=0 \Rightarrow x=-\frac{4}{3}$: $\boxed{\left(-\frac{4}{3},0\right)}$

(ii) $f(0)=\dfrac{4}{-5}$: $\boxed{\left(0,-\frac{4}{5}\right)}$

(b) (i) $\lim\limits_{x\to\pm\infty} f(x)=\dfrac{3}{2}$: $\boxed{y=\frac{3}{2}}$

(ii) $2x-5=0$: $\boxed{x=\frac{5}{2}}$

(c) See sketch →



(d) Let $y=\dfrac{3x+4}{2x-5}$ and swap x and y: $x=\dfrac{3y+4}{2y-5}$

$$x(2y-5)=3y+4 \Rightarrow 2xy-3y=5x+4 \Rightarrow y(2x-3)=5x+4 \Rightarrow \boxed{f^{-1}(x)=\frac{5x+4}{2x-3}}$$

(e) Using the GDC (graph f and f^{-1}, find the intersections):
$\boxed{(-0.45,\ -0.45)}$ and $\boxed{(4.45,\ 4.45)}$ (exactly $x=2\pm\sqrt{6}$, both on $y=x$)

(f) Midpoint: $\left(\dfrac{-0.45+4.45}{2},\dfrac{-0.45+4.45}{2}\right)=(2,2)$.
Both points lie on $y=x$ (gradient 1), so the perpendicular bisector has gradient -1:

$$y-2=-(x-2) \Rightarrow \boxed{y=-x+4}$$

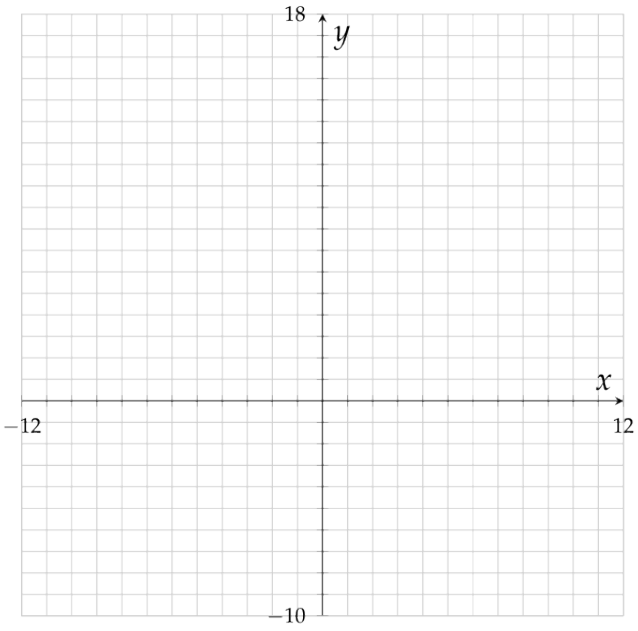
Question 23 Source worksheet · p.39

Question 23 Calculator Medium [Maximum mark: 7]

The function f is given by $f(x) = \dfrac{4x^2 - 128}{x^2 - 16}$, for $x \neq \pm 4$.

(a) Prove that f is an even function. [2]

(b) (i) Sketch the graph of $y = f(x)$. (ii) Write down the range of f. [5]



Answer 23 Source worksheet · p.39

Question 23

(a) For any $x \in \mathbb{R}$, $x \neq \pm 4$:

$$f(-x) = \frac{4(-x)^2 - 128}{(-x)^2 - 16} = \frac{4x^2 - 128}{x^2 - 16} = f(x)$$

Hence f is an even function (its graph is symmetric about the y-axis).

(b) (i) Using the GDC:

V.A. $x = \pm 4$, H.A. $y = 4$

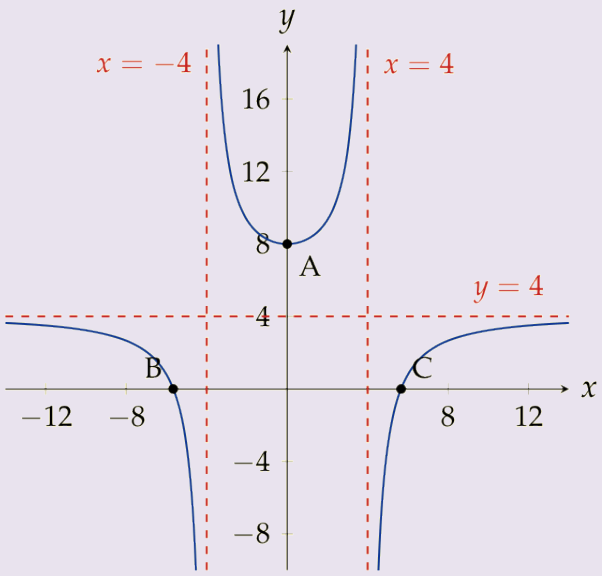
A$(0, 8)$ (local minimum)

B$(-5.66, 0)$, C$(5.66, 0)$

$(x = \pm\sqrt{32} = \pm 4\sqrt{2})$

(ii) Range: $\boxed{y < 4 \text{ or } y \geq 8}$

i.e. $(-\infty, 4) \cup [8, \infty)$



Question 24 Source worksheet · p.41

Question 24 Non-calculator Hard [Maximum mark: 17]

Consider the functions f and g where $f(x) = 3x - 5$ and $g(x) = x - 2$.

(a) Find the inverse function, f^{-1}. [3]

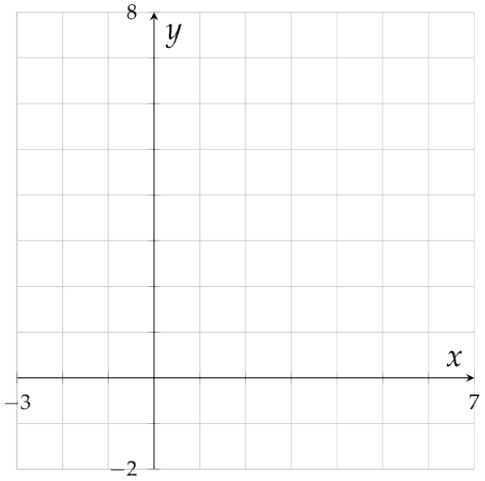
(b) Given that $g^{-1}(x) = x + 2$, find $(g^{-1} \circ f)(x)$. [2]

(c) Given also that $(f^{-1} \circ g)(x) = \frac{x+3}{3}$, solve $(f^{-1} \circ g)(x) = (g^{-1} \circ f)(x)$. [2]

Let $h(x) = \frac{f(x)}{g(x)}$, $x \neq 2$.

(d) (i) Sketch the graph of h for $-3 \leq x \leq 7$ and $-2 \leq y \leq 8$, including any asymptotes.
(ii) Write down the equations of the asymptotes. [5]

(e) (i) Find $h^{-1}(x)$. (ii) Write down the equations of the asymptotes of h^{-1}. [5]



Answer 24 Source worksheet · p.41

Question 24

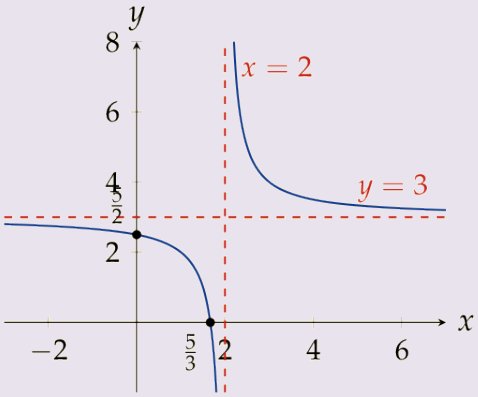
(a) $y = 3x - 5 \Rightarrow x = \dfrac{y+5}{3}$, so $\boxed{f^{-1}(x) = \dfrac{x+5}{3}}$

(b) $(g^{-1} \circ f)(x) = (3x-5) + 2 = \boxed{3x - 3}$

(c) $\dfrac{x+3}{3} = 3x - 3 \Rightarrow x + 3 = 9x - 9 \Rightarrow 8x = 12 \Rightarrow \boxed{x = \frac{3}{2}}$

(d) (i) $h(x) = \dfrac{3x-5}{x-2}$ (sketch →)

(ii) Vertical asymptote $\boxed{x = 2}$, horizontal asymptote $\boxed{y = 3}$



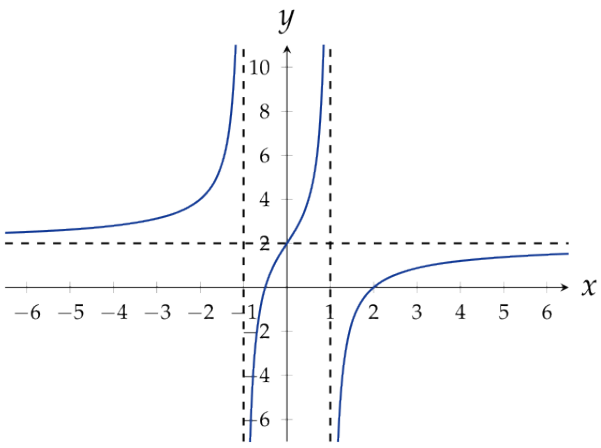
(e) (i) $x = \dfrac{3y-5}{y-2} \Rightarrow xy - 2x = 3y - 5 \Rightarrow y(x-3) = 2x - 5 \Rightarrow \boxed{h^{-1}(x) = \dfrac{2x-5}{x-3}}$

(ii) Vertical asymptote $\boxed{x = 3}$, horizontal asymptote $\boxed{y = 2}$ (the asymptotes of h reflected in $y = x$)

Question 25 Source worksheet · p.43

Question 25 Non-calculator Hard [Maximum mark: 4]

Let $f(x) = p - \dfrac{3x}{x^2 - q^2}$, where $p, q \in \mathbb{R}^+$. Part of the graph of f, including the asymptotes, is shown below.



The equations of the asymptotes are $x = 1$, $x = -1$, $y = 2$. Write down the value of (i) p; (ii) q.

Question 26 Source worksheet · p.43

Question 26 Calculator Hard [Maximum mark: 6]

The function f is defined by $f(x) = \dfrac{3}{\sqrt{9 - x^2}}$, for $-3 < x < 3$.

(a) Sketch the graph of f. [2]

(b) Write down the equation of each vertical asymptote. [2]

(c) Write down the range of the function f. [2]

Answer 25 Source worksheet · p.43

Question 25

(i) As $x \to \pm\infty$, $\dfrac{3x}{x^2 - q^2} \to 0$, so $f(x) \to p$. H.A. $y = 2$ gives $\boxed{p = 2}$.

(ii) Vertical asymptotes where $x^2 - q^2 = 0$, i.e. $x = \pm q$. Since $x = \pm 1$ and $q > 0$: $\boxed{q = 1}$.

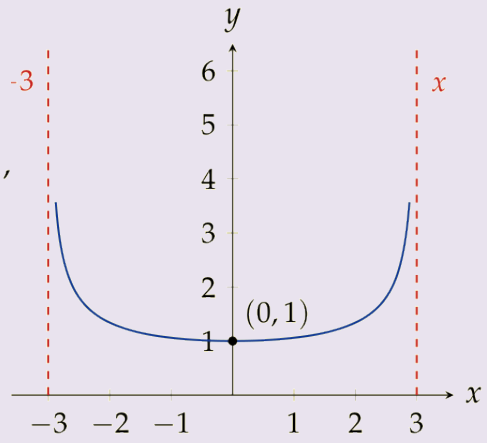
Answer 26 Source worksheet · p.43

Question 26

(a) See sketch → (U-shape, minimum at $(0, 1)$, y-intercept 1)

(b) $9 - x^2 = 0 \Rightarrow$ $\boxed{x = 3}$ and $\boxed{x = -3}$

(c) The minimum value is $f(0) = \frac{3}{3} = 1$, and $f \to \infty$ near $x = \pm 3$: range $\boxed{y \geq 1}$



Question 27 Source worksheet · p.45

Question 27 Calculator Hard [Maximum mark: 7]

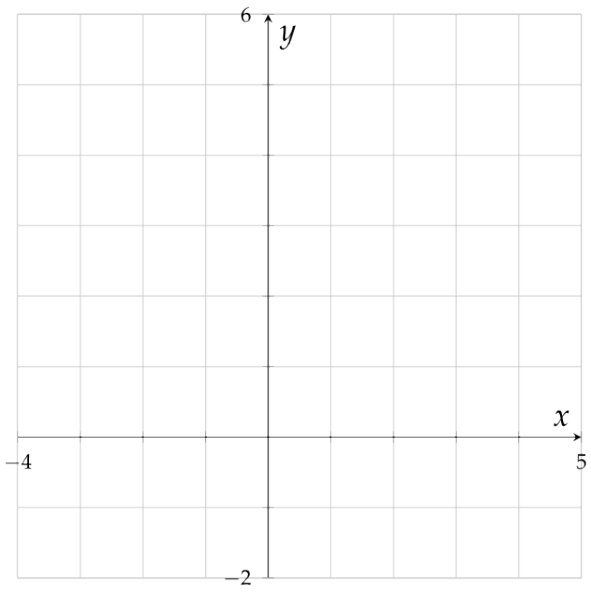
Let $f(x) = \dfrac{4 - ax}{5x - 2}$.

(a) Write down the equation of the vertical asymptote of the graph of $y = f(x)$. [1]

The line $y = -1.4$ is a horizontal asymptote to the graph of $y = f(x)$.

(b) Find the value of a. [2]

(c) The line $y = x - c$, where $c \in \mathbb{R}^+$, intersects the graph of $y = |f(x)|$ at exactly one point. Find the possible values of c. [4]



Answer 27 Source worksheet · p.45

Question 27

(a) $5x - 2 = 0 \Rightarrow \boxed{x = \frac{2}{5}}$

(b) $\lim\limits_{x \to \pm\infty} \dfrac{4 - ax}{5x - 2} = -\dfrac{a}{5} = -1.4 \Rightarrow \boxed{a = 7}$

(c) Graph $y = \left|\dfrac{4 - 7x}{5x - 2}\right|$ on the GDC.

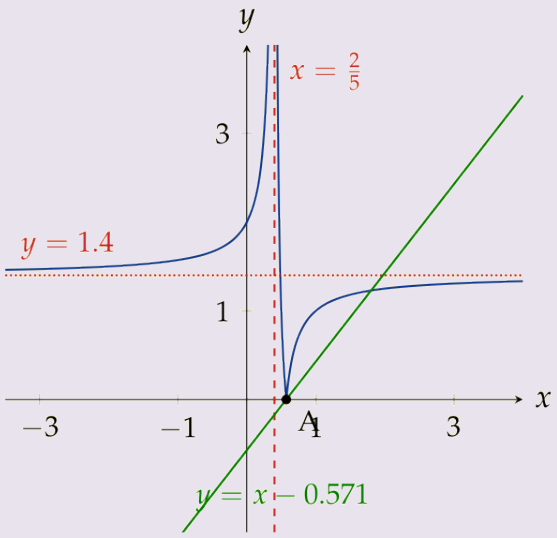
It touches the x-axis at $\mathrm{A}\left(\frac{4}{7}, 0\right) \approx (0.571, 0)$ and has H.A. $y = 1.4$.

$y = x - c$ has gradient 1 and y-intercept $-c < 0$.

The line through A, $y = x - 0.571$, meets the graph at **two** points.

Any line of gradient 1 with y-intercept below -0.571 meets the graph exactly **once**.

Therefore $\boxed{c > 0.571}$ $\left(c > \frac{4}{7}\right)$



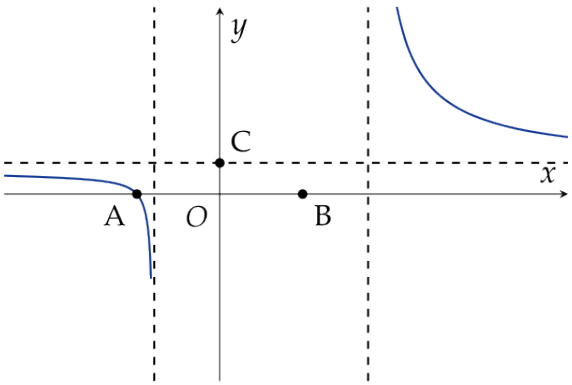
Question 28 Source worksheet · p.47

Question 28 Calculator Hard [Maximum mark: 10]

Let $g(x) = x^4 - 2x^3 + x^2 - 2$.

(a) Solve $g(x) = 0$. [2]

Let $f(x) = \dfrac{2x^3}{g(x)} + 1$. A part of the graph of $f(x)$ is shown below.



(b) Write down (i) the domain of f; (ii) the range of f. [2]

(c) The graph has vertical asymptotes with equations $x = a$ and $x = b$, where $a < b$. Write down the value of a and of b. [2]

(d) The graph has a horizontal asymptote with equation $y = 1$. Explain why the value of $f(x)$ approaches 1 as x becomes very large. [2]

(e) The graph intersects the x-axis at the points A and B. Write down the **exact** value of the x-coordinate at (i) A; (ii) B. [2]

Answer 28

Source worksheet · p.47

(a) $g(x) = x^4 - 2x^3 + x^2 - 2 = x^2(x-1)^2 - 2$. Hence $g(x) = 0$ is equivalent to $x(x-1) = \pm\sqrt{2}$. The negative case gives $x^2 - x + \sqrt{2} = 0$, whose discriminant is $1 - 4\sqrt{2} < 0$; there are no real solutions. For the positive case, $x^2 - x - \sqrt{2} = 0$, so the two real roots are

$$\alpha = \frac{1 - \sqrt{1 + 4\sqrt{2}}}{2} \approx -0.790044, \quad \beta = \frac{1 + \sqrt{1 + 4\sqrt{2}}}{2} \approx 1.790044.$$

(b)(i) Domain of f: $x \in \mathbb{R}$, $x \neq \alpha$, $x \neq \beta$. These are **exact excluded values**, not merely the rounded decimals. (ii) Range: all real values. Between α and β, f is continuous and goes from $+\infty$ to $-\infty$, so it attains every real y-value. (c) The vertical asymptotes are $x = \alpha$ and $x = \beta$; thus $a = \alpha$ and $b = \beta$. (d) As $|x|$ becomes large, $g(x)$ has degree 4 while $2x^3$ has degree 3. Therefore $2\frac{x^3}{g(x)} \to 0$ and $f(x) \to 1$: the horizontal asymptote is $y = 1$. (e) For an x-intercept,

$$f(x) = 0 \Leftrightarrow 2x^3 = -g(x) \Leftrightarrow x^4 + x^2 - 2 = 0 \Leftrightarrow (x^2 + 2)(x^2 - 1) = 0.$$

Only $x = \pm 1$ are real, and neither is a denominator zero. (i) At A: $x = -1$. (ii) At B: $x = 1$. (iii) At C: $f(0) = 1$, so its y-coordinate is 1. **Exact-domain clarification:** use α and β as the excluded values. Decimal approximations are useful for graph labels, but do not replace the exact domain restrictions.